

Advanced Functions

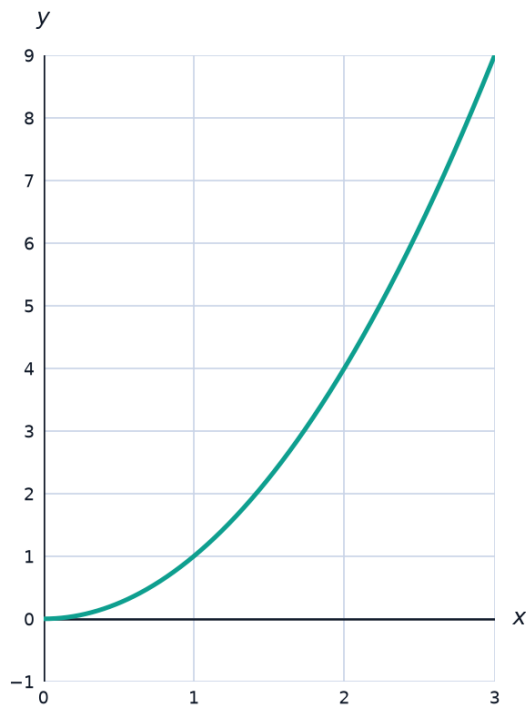


Composite functions with restrictions

- 1 If $f(x) = \sqrt{x}$ and $g(x) = x - 5$, find the domain of $(f \circ g)(x)$.
- 2 If $f(x) = x^2 - 1$ and $g(x) = \frac{1}{x}$, find $(f \circ g)(x)$ and simplify.

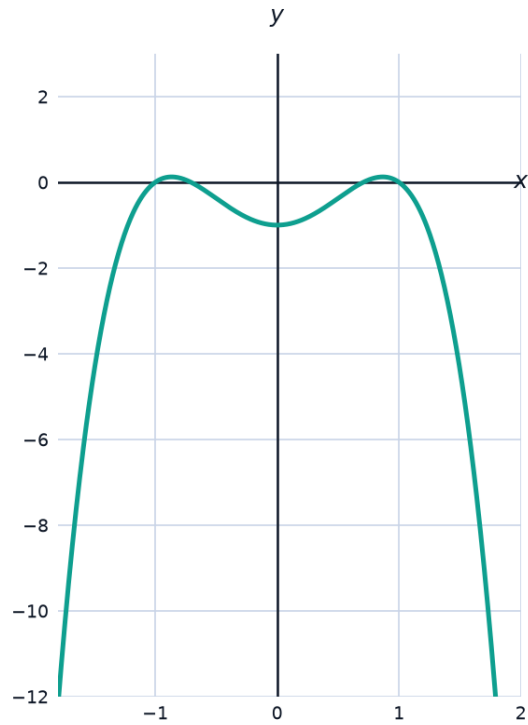
Inverse functions with domain restrictions

- 3 Find the inverse of $f(x) = x^2$ for $x \geq 0$, and explain why the restriction is necessary.

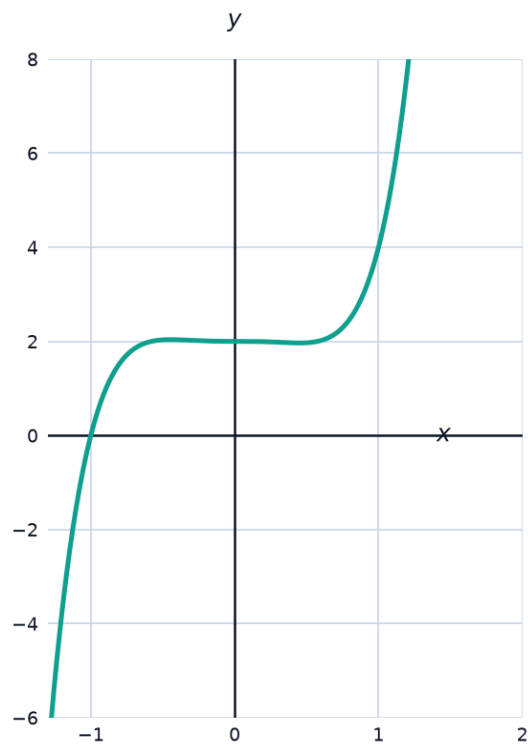


End behavior of polynomial functions

- 4 Describe the end behavior of $f(x) = -2x^4 + 3x^2 - 1$, as $x \rightarrow \infty$ and $x \rightarrow -\infty$.

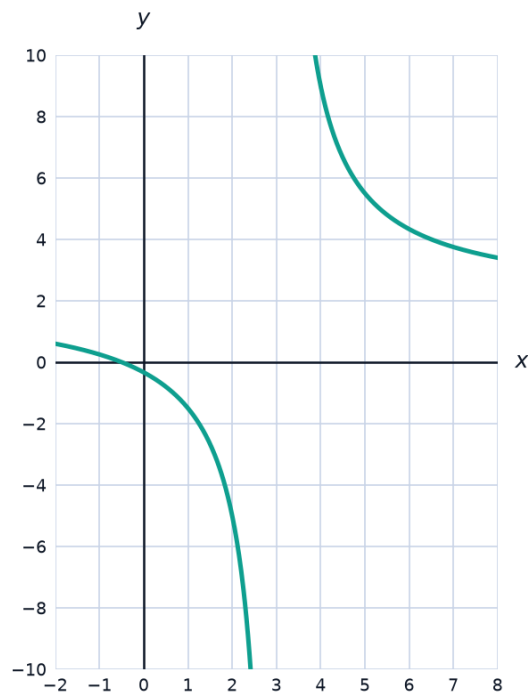


- 5 Describe the end behavior of $g(x) = 3x^5 - x^3 + 2$.



Rational functions: asymptotes and holes

- 6 Find the vertical and horizontal asymptotes of $f(x) = \frac{2x + 1}{x - 3}$.



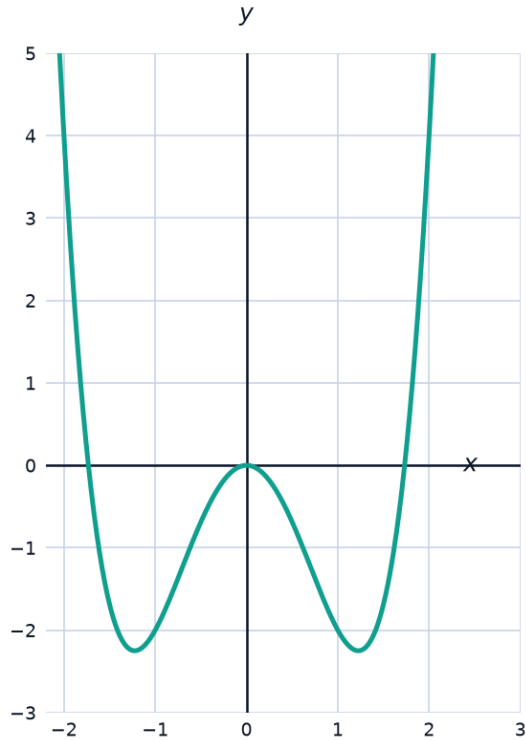
- 7 Find the hole (removable discontinuity) in the graph of $f(x) = \frac{x^2 - 4}{x - 2}$.

Logarithmic functions and their properties

- 8 Simplify $\log_3(9) + \log_3(27)$, using the product rule for logarithms.
- 9 Solve $\ln(x) + \ln(x - 3) = \ln(10)$.

Even, odd, and periodic function classification

- 10 Determine whether $f(x) = x^4 - 3x^2$ is even, odd, or neither.



Piecewise-defined functions and continuity

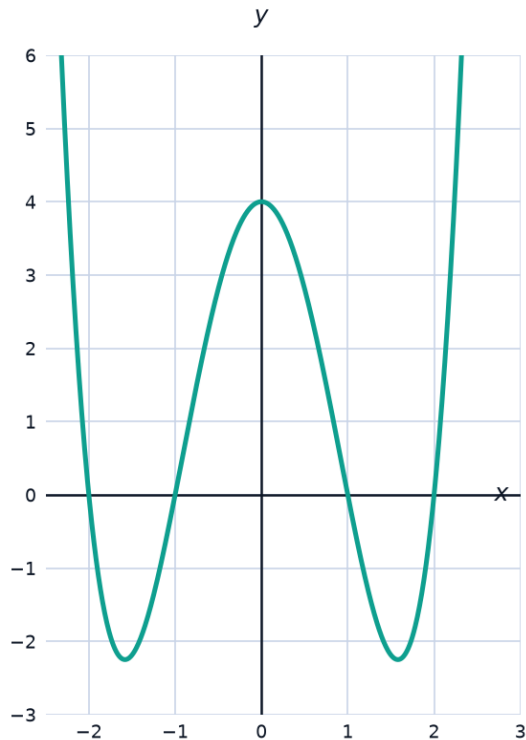
- 11 A function is defined as $f(x) = 3x + 1$ for $x < 2$ and $f(x) = kx - 3$ for $x \geq 2$. Find the value of k that makes f continuous at $x = 2$.

A multi-part function transformation and composition problem

- 12 This question has two parts. Given $f(x) = 2x - 3$ and $g(x) = x^2$. (a) Find $(f \circ g)(x)$ and $(g \circ f)(x)$, in simplified form. (b) Find the value of x for which $(f \circ g)(x) = (g \circ f)(x)$.

Finding zeros of higher-degree functions graphically and algebraically

- 13 A quartic function $f(x) = x^4 - 5x^2 + 4$ can be treated as a quadratic in x^2 . Find all real zeros.

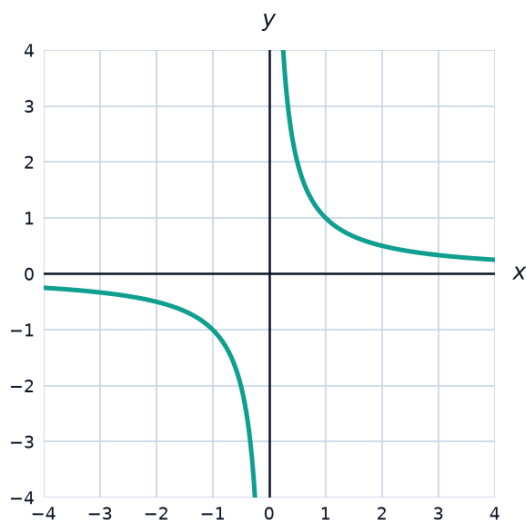


Function operations: sum, difference, product, quotient

- 14 Given $f(x) = x^2 + 1$ and $g(x) = x - 3$, find $(f + g)(x)$, $(f - g)(x)$, and $(fg)(x)$.

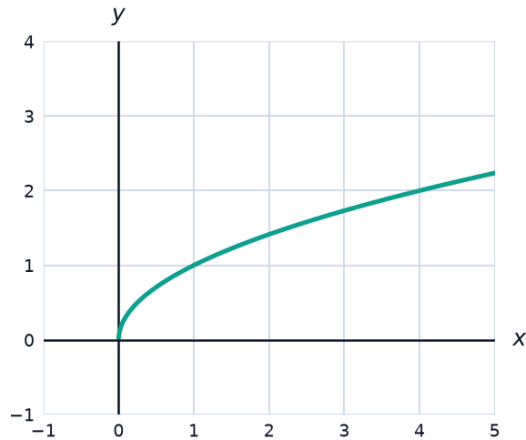
Symmetry and transformations of rational functions

- 15 Explain why $f(x) = \frac{1}{x}$ is symmetric about the origin, and state whether it is even, odd, or neither.



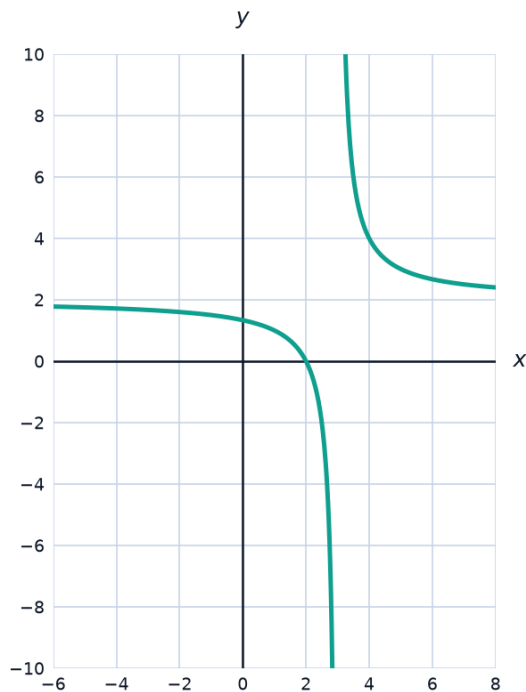
Finding a function's formula from a description of transformations

- 16** The graph of $y = \sqrt{x}$ is shifted 2 units right, stretched vertically by a factor of 3, and shifted 1 unit down. Write the equation of the resulting function.



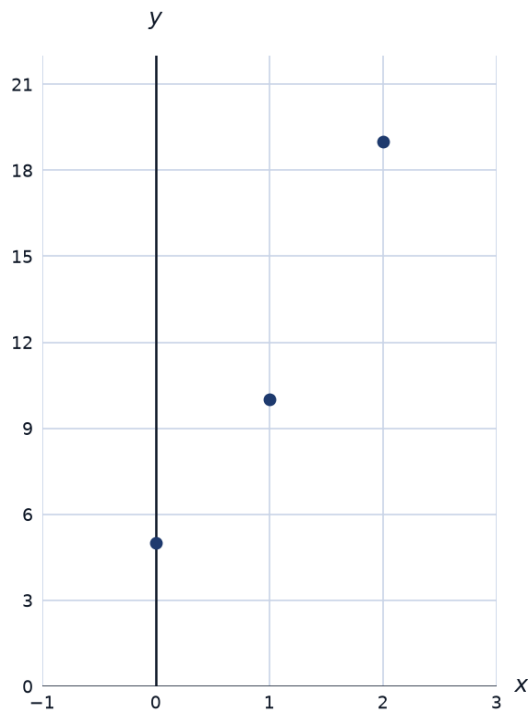
A multi-part rational function analysis problem

- 17** This question has three parts. A function is given by $f(x) = \frac{2x^2 - 8}{x^2 - x - 6}$. (a) Factor the numerator and denominator completely. (b) Identify any holes and vertical asymptotes. (c) State the horizontal asymptote.



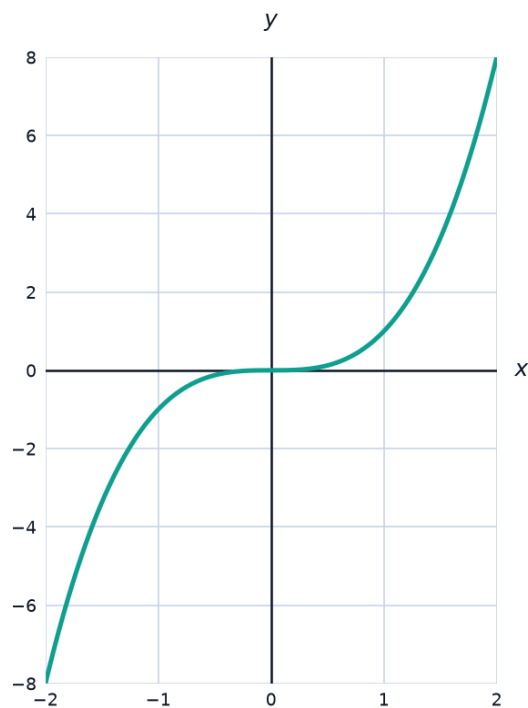
Finding a quadratic function from a table of values

- 18 A quadratic function $f(x) = ax^2 + bx + c$ has $f(0) = 5$, $f(1) = 10$, and $f(2) = 19$. Find a , b , and c .



Determining if a function has an inverse using the horizontal line test

- 19 Explain the horizontal line test, and use it to determine whether $f(x) = x^3$ has an inverse function.



Multiple choice — easy

- 20 Describe the end behavior of $f(x) = 4x^4 - x^3$: as $x \rightarrow \infty$, $f(x) \rightarrow ?$, and as $x \rightarrow -\infty$, $f(x) \rightarrow ?$
- a. $-\infty, \infty$
 - b. $\infty, -\infty$
 - c. $-\infty, -\infty$
 - d. ∞, ∞
- 21 Describe the end behavior of $f(x) = -x^2 - 2x$: as $x \rightarrow \infty$, $f(x) \rightarrow ?$, and as $x \rightarrow -\infty$, $f(x) \rightarrow ?$
- a. ∞, ∞
 - b. $-\infty, \infty$
 - c. $-\infty, -\infty$
 - d. $\infty, -\infty$
- 22 Describe the end behavior of $f(x) = -x^3 + 8x$: as $x \rightarrow \infty$, $f(x) \rightarrow ?$, and as $x \rightarrow -\infty$, $f(x) \rightarrow ?$
- a. $\infty, -\infty$
 - b. $-\infty, \infty$
 - c. ∞, ∞
 - d. $-\infty, -\infty$
- 23 Describe the end behavior of $f(x) = -2x^2 + 9x$: as $x \rightarrow \infty$, $f(x) \rightarrow ?$, and as $x \rightarrow -\infty$, $f(x) \rightarrow ?$
- a. ∞, ∞
 - b. $\infty, -\infty$
 - c. $-\infty, -\infty$
 - d. $-\infty, \infty$

- 24 Describe the end behavior of $f(x) = 4x^2 + 4x$: as $x \rightarrow \infty$, $f(x) \rightarrow ?$, and as $x \rightarrow -\infty$, $f(x) \rightarrow ?$
- a. $\infty, -\infty$
 - b. $-\infty, -\infty$
 - c. $-\infty, \infty$
 - d. ∞, ∞
- 25 Describe the end behavior of $f(x) = -4x^4 - 2x^2$: as $x \rightarrow \infty$, $f(x) \rightarrow ?$, and as $x \rightarrow -\infty$, $f(x) \rightarrow ?$
- a. $-\infty, \infty$
 - b. $\infty, -\infty$
 - c. ∞, ∞
 - d. $-\infty, -\infty$
- 26 Describe the end behavior of $f(x) = 4x^5 + 5x$: as $x \rightarrow \infty$, $f(x) \rightarrow ?$, and as $x \rightarrow -\infty$, $f(x) \rightarrow ?$
- a. $-\infty, -\infty$
 - b. $\infty, -\infty$
 - c. ∞, ∞
 - d. $-\infty, \infty$
- 27 Describe the end behavior of $f(x) = x^2 - 2x$: as $x \rightarrow \infty$, $f(x) \rightarrow ?$, and as $x \rightarrow -\infty$, $f(x) \rightarrow ?$
- a. ∞, ∞
 - b. $-\infty, -\infty$
 - c. $-\infty, \infty$
 - d. $\infty, -\infty$

- 28 Describe the end behavior of $f(x) = x^2 + 9x$: as $x \rightarrow \infty$, $f(x) \rightarrow ?$, and as $x \rightarrow -\infty$, $f(x) \rightarrow ?$
- a. $\infty, -\infty$
 - b. $-\infty, -\infty$
 - c. $-\infty, \infty$
 - d. ∞, ∞
- 29 Describe the end behavior of $f(x) = x^4 + 9x$: as $x \rightarrow \infty$, $f(x) \rightarrow ?$, and as $x \rightarrow -\infty$, $f(x) \rightarrow ?$
- a. $-\infty, \infty$
 - b. ∞, ∞
 - c. $\infty, -\infty$
 - d. $-\infty, -\infty$
- 30 Describe the end behavior of $f(x) = -4x^2 + 4x$: as $x \rightarrow \infty$, $f(x) \rightarrow ?$, and as $x \rightarrow -\infty$, $f(x) \rightarrow ?$
- a. ∞, ∞
 - b. $-\infty, -\infty$
 - c. $\infty, -\infty$
 - d. $-\infty, \infty$
- 31 Describe the end behavior of $f(x) = x^5 + 5x^3$: as $x \rightarrow \infty$, $f(x) \rightarrow ?$, and as $x \rightarrow -\infty$, $f(x) \rightarrow ?$
- a. $-\infty, \infty$
 - b. $\infty, -\infty$
 - c. $-\infty, -\infty$
 - d. ∞, ∞

- 32 Describe the end behavior of $f(x) = x^5 - x^3$: as $x \rightarrow \infty$, $f(x) \rightarrow ?$, and as $x \rightarrow -\infty$, $f(x) \rightarrow ?$
- a. $\infty, -\infty$
 - b. $-\infty, \infty$
 - c. $-\infty, -\infty$
 - d. ∞, ∞
- 33 Describe the end behavior of $f(x) = -3x^2 - 9x$: as $x \rightarrow \infty$, $f(x) \rightarrow ?$, and as $x \rightarrow -\infty$, $f(x) \rightarrow ?$
- a. $-\infty, \infty$
 - b. $\infty, -\infty$
 - c. $-\infty, -\infty$
 - d. ∞, ∞
- 34 Describe the end behavior of $f(x) = -4x^2 + 7x$: as $x \rightarrow \infty$, $f(x) \rightarrow ?$, and as $x \rightarrow -\infty$, $f(x) \rightarrow ?$
- a. $-\infty, -\infty$
 - b. $-\infty, \infty$
 - c. ∞, ∞
 - d. $\infty, -\infty$
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Multiple choice — medium

- 35 Find the vertical asymptote of $f(x) = \frac{x - 15}{x + 14}$.
- a. -13
 - b. -15
 - c. 14
 - d. -14

36 Find the vertical asymptote of $f(x) = \frac{2x - 13}{-x + 10}$.

- a. 9
- b. 11
- c. 10
- d. -10

37 Find the vertical asymptote of $f(x) = \frac{x - 7}{2x - 12}$.

- a. 7
- b. -6
- c. 6
- d. 5

38 Find the vertical asymptote of $f(x) = \frac{2x + 11}{-x - 8}$.

- a. -7
- b. 8
- c. -9
- d. -8

39 Find the vertical asymptote of $f(x) = \frac{7x + 1}{3x + 6}$.

- a. 2
- b. -3
- c. -2
- d. -1

40 Find the vertical asymptote of $f(x) = \frac{-8x - 1}{7x + 7}$.

- a. 1
- b. -1
- c. 0
- d. -2

41 Find the vertical asymptote of $f(x) = \frac{8x - 7}{-5x - 5}$.

- a. -2
- b. 0
- c. -1
- d. 1

42 Find the vertical asymptote of $f(x) = \frac{-8x + 1}{x}$.

- a. -1
- b. 0
- c. 1
- d. 2

43 Find the vertical asymptote of $f(x) = \frac{9x - 10}{x - 7}$.

- a. -7
- b. 7
- c. 6
- d. 8

44 Find the vertical asymptote of $f(x) = \frac{-2x + 6}{3x}$.

- a. 0
- b. -2
- c. -1
- d. 1

45 Find the vertical asymptote of $f(x) = \frac{5x + 4}{-5x - 15}$.

- a. -4
- b. 3
- c. -3
- d. -2

46 Find the vertical asymptote of $f(x) = \frac{-9x - 12}{-x - 6}$.

- a. -7
- b. -5
- c. 6
- d. -6

47 Find the vertical asymptote of $f(x) = \frac{-7x + 14}{-2x + 14}$.

- a. 8
- b. 7
- c. -7
- d. 6

48 Find the vertical asymptote of $f(x) = \frac{x - 12}{7x - 14}$.

- a. -2
- b. 2
- c. 3
- d. 1

49 Find the vertical asymptote of $f(x) = \frac{7x + 14}{3x - 9}$.

- a. 2
- b. 4
- c. -3
- d. 3

Multiple choice — hard

50 Solve $\ln(x) + \ln(x - 5) = \ln(55)$.

- a. 11
- b. 5
- c. 10
- d. 12

51 Solve $\ln(x) + \ln(x - 1) = \ln(5)$.

- a. 4
- b. 1
- c. 6
- d. 5

52 Solve $\ln(x) + \ln(x - 9) = \ln(72)$.

a. 9

b. 6

c. 8

d. 7

53 Solve $\ln(x) + \ln(x - 8) = \ln(32)$.

a. 3

b. 8

c. 4

d. 5

54 Solve $\ln(x) + \ln(x - 1) = \ln(7)$.

a. 8

b. 6

c. 1

d. 7

55 The quartic function $f(x) = x^4 - 5x^2 + 4$ can be treated as a quadratic in x^2 . Find all real zeros.

a. $-2, -1, 1, 2$

b. $-1, -2, 1, 2$

c. $1, 2$

d. $-2, 2$

- 56 The quartic function $f(x) = x^4 - 25x^2 + 144$ can be treated as a quadratic in x^2 . Find all real zeros.
- a. 3, 4
 - b. -4, 4
 - c. -3, -4, 3, 4
 - d. -4, -3, 3, 4
- 57 The quartic function $f(x) = x^4 - 13x^2 + 36$ can be treated as a quadratic in x^2 . Find all real zeros.
- a. 2, 3
 - b. -2, -3, 2, 3
 - c. -3, -2, 2, 3
 - d. -3, 3
- 58 The quartic function $f(x) = x^4 - 17x^2 + 16$ can be treated as a quadratic in x^2 . Find all real zeros.
- a. -1, -4, 1, 4
 - b. 1, 4
 - c. -4, -1, 1, 4
 - d. -4, 4
- 59 The quartic function $f(x) = x^4 - 20x^2 + 64$ can be treated as a quadratic in x^2 . Find all real zeros.
- a. -4, 4
 - b. -2, -4, 2, 4
 - c. -4, -2, 2, 4
 - d. 2, 4

60 If $f(x) = -3x + 5$ and $g(x) = x^2 + 5$, find $(f \circ g)(3)$.

a. -39

b. -37

c. -35

d. 14

61 If $f(x) = 3x + 6$ and $g(x) = x^2 + 4$, find $(f \circ g)(5)$.

a. 29

b. 93

c. 91

d. 95

62 If $f(x) = -3x - 3$ and $g(x) = x^2 - 3$, find $(f \circ g)(1)$.

a. 5

b. 1

c. 3

d. -2

63 If $f(x) = 2x - 1$ and $g(x) = -2x^2$, find $(f \circ g)(-1)$.

a. -7

b. -2

c. -3

d. -5

64 If $f(x) = -2x + 4$ and $g(x) = 3x^2 - 4$, find $(f \circ g)(-5)$.

a. 71

b. -136

c. -138

d. -140